

SION AP, Switzerland

WMO: 067200

Lat: 46.2186N

Lon: 7.3303E

Elev: 482

StdP: 95.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-6.5	-12.1	1.4	-4.4	-10.3	1.6	-3.4	8.1	5.1	6.8	4.7	1.5	80	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	31.7	19.8	29.8	19.2	28.1	18.6	20.6	29.7	19.8	28.2	19.1	26.7	5.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	13.4	22.7	17.0	12.8	22.4	16.3	12.3	22.1	61.5	29.9	58.7	28.3	56.2	26.9	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.0	6.9	-10.0	34.5	2.4	2.0	-11.7	36.0	-13.1	37.2	-14.5	38.3	-16.2	39.8		
(5)			-10.5	22.1	2.3	0.7	-12.2	22.6	-13.5	23.0	-14.8	23.4	-16.5	23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	1.0	3.1	7.7	11.6	15.6	19.5	21.0	20.3	16.2	11.8	5.7	1.4		
	DBStd	7.83	3.41	3.63	3.55	3.45	3.18	3.40	3.06	3.07	3.07	3.20	3.80	3.37		
	HDD10.0	1004	278	194	87	23	2	0	0	1	19	136	266			
	HDD18.3	2826	536	427	328	202	94	27	11	16	77	203	380	524		
	CDD10.0	1474	0	0	17	71	177	283	342	318	187	74	6	0		
	CDD18.3	256	0	0	0	1	11	61	94	75	14	0	0	0		
	CDH23.3	2306	0	0	0	21	136	576	843	624	104	2	0	0		
	CDH26.7	713	0	0	0	1	22	183	288	205	13	0	0	0		
(14) Wind	WSAvg	2.2	1.5	1.8	2.4	2.7	2.7	2.8	2.7	2.4	2.3	1.7	1.5	1.4		
(15) Precipitation	PrecAvg	819	62	62	54	59	78	76	83	79	60	61	72	74		
	PrecMax	1112	185	229	152	109	154	146	169	156	144	155	187	205		
	PrecMin	594	10	5	20	12	21	35	40	28	18	4	6	0		
	PrecStd	124	45	54	34	27	42	30	30	32	31	35	43	48		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	15.4	21.0	25.9	29.0	33.2	34.1	34.0	28.6	23.5	17.7	12.1		
		MCWB	6.7	7.6	10.7	14.0	17.5	20.3	20.3	20.7	18.2	14.6	10.8	7.4		
	2%	DB	8.9	12.8	18.8	23.1	26.7	30.8	31.8	31.1	26.1	21.1	15.2	9.1		
		MCWB	5.2	6.3	10.0	12.7	16.2	19.5	19.8	19.8	17.4	14.1	9.8	6.2		
	5%	DB	7.1	10.6	16.6	21.0	24.6	28.8	29.7	28.8	24.2	19.2	13.0	7.4		
		MCWB	4.4	5.3	9.0	11.8	15.2	18.6	19.3	19.2	16.7	13.2	8.7	5.2		
	10%	DB	5.6	8.5	14.3	18.8	22.6	26.8	27.9	26.9	22.4	17.3	11.1	6.1		
		MCWB	3.4	4.6	7.9	10.7	14.3	17.7	18.7	18.5	15.7	12.4	8.0	4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	8.5	11.7	14.7	18.4	21.4	21.6	21.5	19.2	16.2	11.7	8.1		
		MCDB	10.2	13.2	19.1	24.1	27.2	31.3	31.2	31.2	25.9	20.3	15.3	10.7		
	2%	WB	6.0	7.1	10.5	13.3	17.0	20.1	20.6	20.5	18.1	15.1	10.4	6.7		
		MCDB	8.0	11.6	17.3	21.7	25.0	29.2	29.8	29.3	24.6	19.1	13.7	8.5		
	5%	WB	4.8	5.9	9.5	12.2	15.9	19.1	19.8	19.6	17.2	14.1	9.4	5.7		
		MCDB	6.7	9.7	15.3	19.7	23.0	27.3	28.3	27.3	23.0	17.8	12.4	7.2		
	10%	WB	3.7	4.7	8.4	11.2	14.9	18.2	19.1	18.9	16.3	13.1	8.3	4.5		
		MCDB	5.3	8.0	13.5	18.0	21.3	25.4	26.8	25.6	21.3	16.6	10.9	6.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	9.0	11.1	11.7	11.6	11.8	12.0	11.4	10.8	10.2	8.3	6.7		
		MCDBR	8.5	12.4	14.9	15.4	15.0	14.8	14.9	14.9	13.1	12.3	10.7	7.9		
		MCWBR	5.5	7.0	7.6	7.1	6.7	6.1	5.7	6.1	6.0	6.4	6.1	5.3		
		MCWBR	7.5	11.1	13.4	14.2	13.5	13.5	13.6	13.3	11.8	10.7	9.4	6.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.271	0.300	0.352	0.398	0.419	0.423	0.419	0.405	0.378	0.350	0.307	0.271		
		taud	2.305	2.231	2.149	2.136	2.174	2.224	2.234	2.285	2.343	2.392	2.402	2.347		
		Ebn at Noon	829	868	865	855	849	846	843	841	833	804	780	788		
		Edh at Noon	76	101	129	143	143	137	134	123	106	87	70	66		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.46	2.33	3.44	4.38	5.00	5.95	5.93	5.17	4.05	2.62	1.50	1.20		
		RadStd	0.17	0.32	0.47	0.59	0.43	0.53	0.49	0.43	0.34	0.28	0.21	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.52	N/A	N/A	+1.33	+0.29	N/A	N/A	N/A	+145	+72
(47) Regional (105 neighbors)		N/A	N/A	N/A	+0.71	+0.38	+0.29	N/A	N/A	+55	+17

Nomenclature: See separate page